

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 17¹****Topics covered :**

1. Method of variation of parameters

- The [method of variation of parameters](#) is a technique used to find a particular solution to a nonhomogeneous linear differential equation by leveraging the solutions to the corresponding homogeneous equation, which are often simpler to determine. Consider the following second-order nonhomogeneous linear differential equation

$$y'' + p(x)y' + q(x)y = g(x) \quad (1)$$

Let the general solution to the associated homogeneous equation be written as

$$y_c = c_1 y_1(x) + c_2 y_2(x), \quad (2)$$

where $y_1(x)$ and $y_2(x)$ are linearly independent, and c_1 and c_2 are arbitrary constants. The method of variation of parameters proposes that a particular solution to the original nonhomogeneous equation can be expressed as

$$y_p(x) = v_1(x)y_1(x) + v_2(x)y_2(x) \quad (3)$$

where $v_1(x)$ and $v_2(x)$ are unknown functions that need to be determined. In order to determine, $v_1(x)$ and $v_2(x)$, we need two conditions, which are chosen as follows. Since $y_p(x)$ must satisfy (1), we compute the first and second derivatives of $y_p(x)$. Using the product rule, we have

$$y'_p(x) = v'_1(x)y_1(x) + v_1(x)y'_1(x) + v'_2(x)y_2(x) + v_2(x)y'_2(x), \quad (4)$$

and for the second derivative

$$y''_p(x) = v''_1(x)y_1(x) + 2v'_1(x)y'_1(x) + v_1(x)y''_1(x) + v''_2(x)y_2(x) + 2v'_2(x)y'_2(x) + v_2(x)y''_2(x). \quad (5)$$

Substituting these expressions for $y'_p(x)$ and $y''_p(x)$ in (1) results in

$$\frac{d}{dx} [v'_1(x)y_1(x) + v'_2(x)y_2(x)] + p(x) [v_1(x)y_1(x) + v_2(x)y_2(x)] + v'_1(x)y'_1(x) + v'_2(x)y'_2(x) = g(x). \quad (6)$$

To simplify the problem, we avoid the extra complexity of second derivatives of $v_1(x)$ and $v_2(x)$ by making the assumption (which gives the first condition to determine $v_1(x)$ and $v_2(x)$)

$$v'_1(x)y_1(x) + v'_2(x)y_2(x) = 0. \quad (7)$$

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This assumption eliminates the second derivatives of $v_1(x)$ and $v_2(x)$, which simplifies the equations. Substituting (7) in (6), and simplifying results in

$$v_1'(x)y_1'(x) + v_2'(x)y_2'(x) = g(x) \quad (8)$$

Equations (7) and (8) indicate the two conditions that can be used to compute the derivatives of the two unknown functions $v_1(x)$ and $v_2(x)$. Using Cramer's rule, the solutions for $v_1'(x)$ and $v_2'(x)$ are given by

$$v_1'(x) = \frac{W_1}{W} = \frac{-y_2(x)g(x)}{W} \text{ and } v_2'(x) = \frac{W_2}{W} = \frac{y_1(x)g(x)}{W} \quad (9)$$

where W is the Wronskian. Thus, we can determine $v_1(x)$ and $v_2(x)$ by integrating $v_1'(x)$ and $v_2'(x)$ with respect to x i.e.,

$$v_1(x) = \int v_1'(x) dx, \quad v_2(x) = \int v_2'(x) dx. \quad (10)$$

Upon computing $v_1(x)$ and $v_2(x)$, we can substitute these functions in (3) to obtain the particular solution $y_p(x)$ i.e.,

$$y_p(x) = v_1(x)y_1(x) + v_2(x)y_2(x). \quad (11)$$

This $y_p(x)$ is the particular solution to the nonhomogeneous differential equation.